

SENTINELLAB

Complete User Guide & Operator Handbook

Local SOC training and security-operations platform | Edition 1 | 10 October 2026

Covers all 15 sidebar modules, global controls, workflows, administration, troubleshooting, and safety limitations. Based on the feature matrix, developer history, Docker guide, and current frontend code on the `fix/phase1-security-regressions` branch.

Important: Developer preview, not a production-certified SIEM. Some actions are simulated, and certain integrations are unavailable in the portable Docker build.

1. Read this first

SentinelLab is a security operations center (SOC) learning and investigation environment. It collects and normalizes telemetry, evaluates detection rules, presents alerts, supports investigation cases, and includes testing, threat intelligence, MITRE coverage, playbooks, reports and administrative controls.

Feature maturity and safety

- Core API authentication, workspace isolation, selected detection functionality and Docker smoke flows have automated test coverage; this does not mean every UI workflow has been browser-tested.
- Training Lab is for synthetic/demo telemetry. Production workspace access is privileged and governed by additional approvals.
- External playbook actions (EDR isolation, firewall block, identity disable, notifications) are recorded/simulated when no live external system is connected; they do not actually change external systems.
- Portable Docker build does not bundle the Emergent AI adapter. AI assistant availability depends on deployment configuration.
- This manual describes UI capabilities seen in repository source; controls and behavior may evolve.

2. Install and open SentinelLab

1. Install Docker Desktop for Windows and start the Docker Engine; verify with docker info.
2. Clone the SentinelLab repository and check out branch fix/phase1-security-regressions.
3. Copy .env.example to .env and configure strong, unique JWT_SECRET and ADMIN_PASSWORD. Do not commit .env.
4. From the repository root run: docker compose up -d --build --wait.
5. Open <http://127.0.0.1:8080> (or port 8081 if you configured a local port override).
6. Sign in using the configured admin email and password. API documentation is at <http://127.0.0.1:8000/docs> if the API port is published.

If Docker Hub times out

For the current PowerShell session, set MONGO_IMAGE=public.ecr.aws/docker/library/mongo:7, PYTHON_BASE_IMAGE=public.ecr.aws/docker/library/python:3.12-slim, NODE_BASE_IMAGE=public.ecr.aws/docker/library/node:22-alpine, and NGINX_BASE_IMAGE=public.ecr.aws/docker/library/nginx:1.27-alpine, using \$env:VARIABLE = "value"; then retry Compose.

Maintenance commands

- docker compose ps — check running containers.
- docker compose logs --tail=100 api — inspect backend logs.
- docker compose down — stop containers while retaining database volume.
- Never run docker compose down -v unless you intentionally want to delete the MongoDB volume.

3. Navigation and shared controls

1. Use the left sidebar to open any of the 15 modules described below.

2. Use the top global search field to find events, alerts, investigations, rules and assets; results link to the relevant module.
3. Use the workspace selector to switch between Training Lab and authorized production workspaces; the view reloads.
4. Choose a top-bar time range: 15m, 1h, 24h, 7d or 30d; note that some pages have their own filters.
5. Open the notification bell to inspect recent notices; opening it marks notifications read.
6. Use the palette control to switch themes; use the account menu to sign out.

4. Overview dashboard

High-level SOC posture, key metrics, recent alert activity and visual summaries based on current workspace and time range.

How to use it

1. Select the desired workspace and time range.
2. Review key performance indicators and charts for volume, severity and alert trends.
3. Open recent alerts or navigate to Alerts for triage.

What to know

- Training Lab may display synthetic records. Dashboard values depend on ingested records; an empty dashboard does not prove a system is secure.

5. Events Explorer

Searchable event store for parsed, normalized and raw telemetry.

How to use it

1. Select severity and event-type filters and adjust other available filter inputs.
2. Review the paginated events table: timestamp, severity, type, source, host, source IP, user and action.
3. Open an event to inspect overview, raw, normalized and related-data tabs.
4. Use Copy Event ID or Copy JSON where shown; export filtered results as CSV or JSON.
5. Use Previous/Next to browse result pages.

What to know

- Event fields depend on the original source and parser. Validate timestamps and normalization before using data as evidence.

6. Threat Hunting

Interactive hunts over stored telemetry with query templates, facets and timelines.

How to use it

1. Open Threat Hunting and select a suggested template or enter a supported hunt query.
2. Run the query and inspect the returned event list and summary facets.

3. Narrow the search with supported filters; save useful hunts for reuse when offered.
4. Pivot into Events Explorer or investigations to follow suspicious activity.

What to know

- The query parser is intentionally constrained; this is not an unrestricted database query interface.

7. Alerts

Triage detections generated by rules or indicator scans.

How to use it

1. Filter alerts by status and severity.
2. Open an alert and review its severity, title, rule, host and supporting events.
3. Update status, assign a responder and add comments as appropriate.
4. Investigate linked events and create or open an investigation case.

What to know

- Alert details are scoped to the selected workspace. Changes may be audited.

8. Detection Rules

Manage rules that match, count or correlate telemetry and trigger alerts.

How to use it

1. Review rules, their type, severity, enabled state, match count and last run.
2. Open a rule to inspect its configuration and linked MITRE technique mappings.
3. Use Create Rule to enter rule metadata, conditions and supported match settings.
4. Toggle rules on/off deliberately; use Backtest (7d) to evaluate against historical data.
5. Use Detection Replay for isolated regression checks before changing important rules.

What to know

- Built-in detection rules are supplied; changing a rule can change alert volume. Backtest and replay are distinct from live ingestion.

9. Detection Replay

An isolated test bench that evaluates sample or uploaded telemetry against enabled detection rules.

How to use it

1. Load one of the built-in sample scenarios or provide telemetry in a supported format (JSON, JSONL, CSV, Syslog or CEF).
2. Choose a format and configure expected detections if the UI offers them.
3. Run replay and compare actual matches with expected matches.

4. Inspect matched, missing and unexpected results to tune detection logic.

What to know

- Replay is designed not to persist events or alerts. It is a testing tool, not a live ingestion channel.

10. MITRE ATT&CK

Coverage explorer that maps enabled rules and alerts onto a curated subset of ATT&CK techniques.

How to use it

1. Review the tactic/technique coverage grid.
2. Select a technique to inspect mapped rules and alert counts.
3. Review gaps and decide whether new detections are warranted.
4. Use Detection Rules to implement and test a rule for a meaningful gap.

What to know

- Coverage reflects a curated subset (12 tactics / 36 techniques in development history), not the entire ATT&CK catalog. Coverage is not a measure of guaranteed protection.

11. Sources & Ingestion

Configure telemetry sources, issue ingestion tokens and upload supported log files.

How to use it

1. Click Add Source; provide its name, source type and host/endpoint fields.
2. Create the source and copy its generated token immediately; it is displayed once.
3. Use the source upload control to import telemetry files in supported formats.
4. Check source event counts, last-received time and status; delete unused sources carefully.
5. Consult the API docs for programmatic ingestion routes and authorization requirements.

What to know

- Supported parser formats include JSON, JSONL, CSV, Syslog RFC 3164/5424 and CEF. Store source tokens securely.

12. Pipeline Observatory

Read-only telemetry pipeline monitoring for ingestion volume and parsing quality.

How to use it

1. Review current events-per-second and indexed event counts.
2. Inspect the recent throughput series and per-source throughput/health table.
3. Check parser failure counts and recent parse errors.
4. Use Sources & Ingestion to correct malformed or disconnected inputs.

What to know

- The current pipeline is synchronous; retry/backlog counters may be zero by design, not proof of an enterprise queue.

13. Threat Intel

Manage indicators of compromise (IOCs), import feeds and correlate with stored events.

How to use it

1. Review indicator counts and filter by type/status.
2. Click Add Indicator to enter type, value, confidence, source and tags.
3. Use Import Indicators for CSV (type,value,confidence,source,tags) or STIX 2.1 bundle JSON.
4. Inspect IOC hit details to find matching events; use indicator scanning when available to evaluate stored telemetry.
5. Mark known false positives or delete incorrect indicators with care.

What to know

- Supported types include IP, domain, URL, hash and email. IOC confidence is contextual, not an automatic verdict.

14. Investigations

Case management for alert triage, analyst collaboration and evidence tracking.

How to use it

1. Click New Investigation and enter a case title, severity and other requested fields.
2. Open a case to review related alerts/events and assigned lead.
3. Add notes and response tasks; update case status and assignments.
4. Attach evidence where available and retain hashes/chain-of-custody details.
5. Use Investigation Graph to explore relationships around relevant entities.

What to know

- Evidence hashing supports integrity checks; it does not replace organizational evidence-handling policy.

15. Investigation Graph

Interactive node-link relationships derived from stored events, alerts, investigations and IOC matches.

How to use it

1. Select a seed type such as alert, IP, host, user or indicator.
2. Enter the seed identifier and build the graph.
3. Drag nodes to rearrange; click a node to inspect its metadata.
4. Use Expand to pivot into neighbors and follow related entities.

What to know

- The visualization is capped for readability (70 nodes noted in development history). Missing nodes do not necessarily imply no relationship.

16. Incident Response Playbooks

Seven built-in analyst-controlled workflows with dry-run, execution history and explicit approval gates.

How to use it

1. Open Playbooks and select a workflow such as Malware Alert Triage.
2. Click Run Playbook; choose a Target Alert or Manual (no alert).
3. Review the listed steps and any approval-gated external actions.
4. Use Dry Run first: inspect the step-by-step result without making persisted changes.
5. Use Execute Live only when appropriate; approved local steps may create cases and tasks.
6. Open Execution History and inspect each step status and details.

What to know

- A manual run may have no related events or indicators; enrichment should be skipped when no IOC is available.
- Approval for EDR isolation/firewall block/identity disable/notification does not mean the external action occurred; with no connector, the action is simulated or audit-only.
- Statuses include ok, skipped, simulated, pending_approval and error. A live run can still create investigations/tasks even if another step fails.
- The UI currently says Execute Live; interpret this as executing local workflow effects, not necessarily connected external actions.

17. Reports

Generate and download SOC reports from workspace data.

How to use it

1. Choose a report template and time range.
2. Generate a report, then inspect the report history table.
3. Download the generated report as JSON or CSV.
4. Review scope and date range before sharing the export.

What to know

- Native PDF report export and scheduled reports are not implemented in the documented version. This separate user guide is a PDF, not an in-app report export.

18. Settings

Administration and user preferences, divided into six tabs.

How to use it

1. General: configure application name, time zone, default time range, theme and demo mode; reset Training Lab synthetic data only when intentional.
2. Security: review security controls; authorized admins can manage users/roles. Super admins can review production membership and approval requests.
3. Ingestion: inspect pipeline stages, supported formats and parser/schema version.
4. Retention: review and adjust the retention controls available in your deployment.
5. Notifications: configure available notification preferences.
6. Audit: inspect available security/audit history.

What to know

- Production access requires a request and approval by a different super admin. A requester cannot approve their own request.
- Interrupted approvals may be reconciled only after the five-minute safety window; reconciliation checks actual membership and does not itself grant access.
- Some settings are informational displays rather than toggles. Role and workspace controls depend on privileges.

19. Suggested learning walkthrough

1. Log in to Training Lab, choose a 24-hour time range, and inspect Overview.
2. Open Sources & Ingestion and review existing source health.
3. Use Events Explorer to inspect a synthetic event and its raw/normalized fields.
4. Open Alerts, select a detection, and review related events.
5. Open Detection Rules and inspect the corresponding rule and backtest behavior.
6. Run a Detection Replay sample; verify expected-vs-actual outcomes.
7. Open MITRE ATT&CK to see how that rule contributes to coverage.
8. Create an Investigation, add a note and task, then explore its Graph.
9. Inspect Threat Intel IOC hits and the Pipeline Observatory.
10. Run a Playbook in Dry Run mode first, then read each step status.
11. Generate a Report and download CSV/JSON.
12. Visit Settings and review role, membership and audit safeguards.

20. Troubleshooting and FAQ

Docker command not recognized

Install Docker Desktop or reopen PowerShell after installation; verify `docker --version`.

Docker Engine API pipe missing

Start Docker Desktop and wait for the engine to become ready; rerun `docker info`.

Docker Hub OAuth timeout

Use the public ECR mirror image variables listed in Section 2; check network connectivity.

Port 8080 already allocated

Use a Compose override to publish the web container on a different localhost port, such as 8081.

No events or alerts

Check workspace, time range, source health and ingestion. Training Lab data may need a deliberate reset.

AI assistant unavailable

The portable Docker runtime does not bundle the Emergent AI adapter; check your deployment integrations.

EDR isolation shows simulated

No real EDR connector is attached. Acknowledging/approving a playbook step does not isolate a host.

Manual enrichment skipped

A manual playbook without a target alert may have no source IP/IOC; supply a relevant alert for meaningful enrichment.

Permission denied

Check your assigned role, workspace membership and production approval status.

Where are API details?

Open /docs on the local API port if published; API paths and behavior may vary by deployment.

21. Terminology

- SOC: Security Operations Center; SIEM: Security Information and Event Management.
- IOC: Indicator of Compromise, e.g. suspicious IP, domain or hash.
- EDR: Endpoint Detection and Response; requires a connected endpoint platform for actual isolation.
- RBAC: Role-Based Access Control; permissions depend on user role.
- MITRE ATT&CK: Knowledge base of adversary tactics and techniques.
- Replay: Non-persistent detection test against selected telemetry.
- Dry Run: Preview/test without applying workflow changes.
- Audit Trail: Recorded activity used for accountability and review.

22. Reference and publication

Repository: github.com/ibrahim1101/SentinelLab. Documentation source: docs/FEATURE_MATRIX.md, docs/DEVELOPMENT_HISTORY.md, docs/LOCAL_DOCKER.md, frontend/src/pages/*.jsx and backend/playbooks.py, reviewed on 10 October 2026.

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Version note: this is a source-based operator manual, not a claim that every control was individually verified in a browser. Update the guide when UI labels, security behavior or external integrations change.